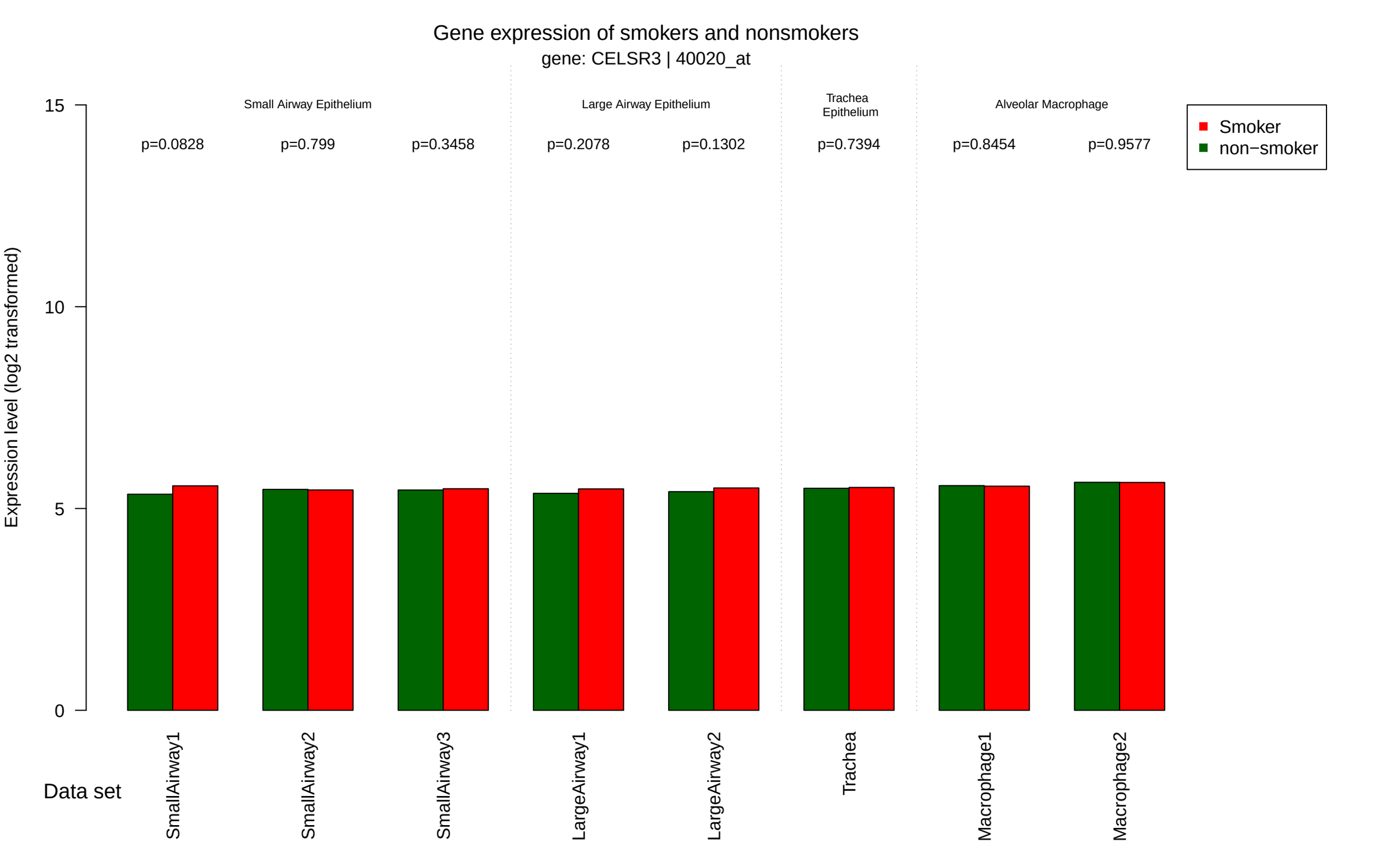
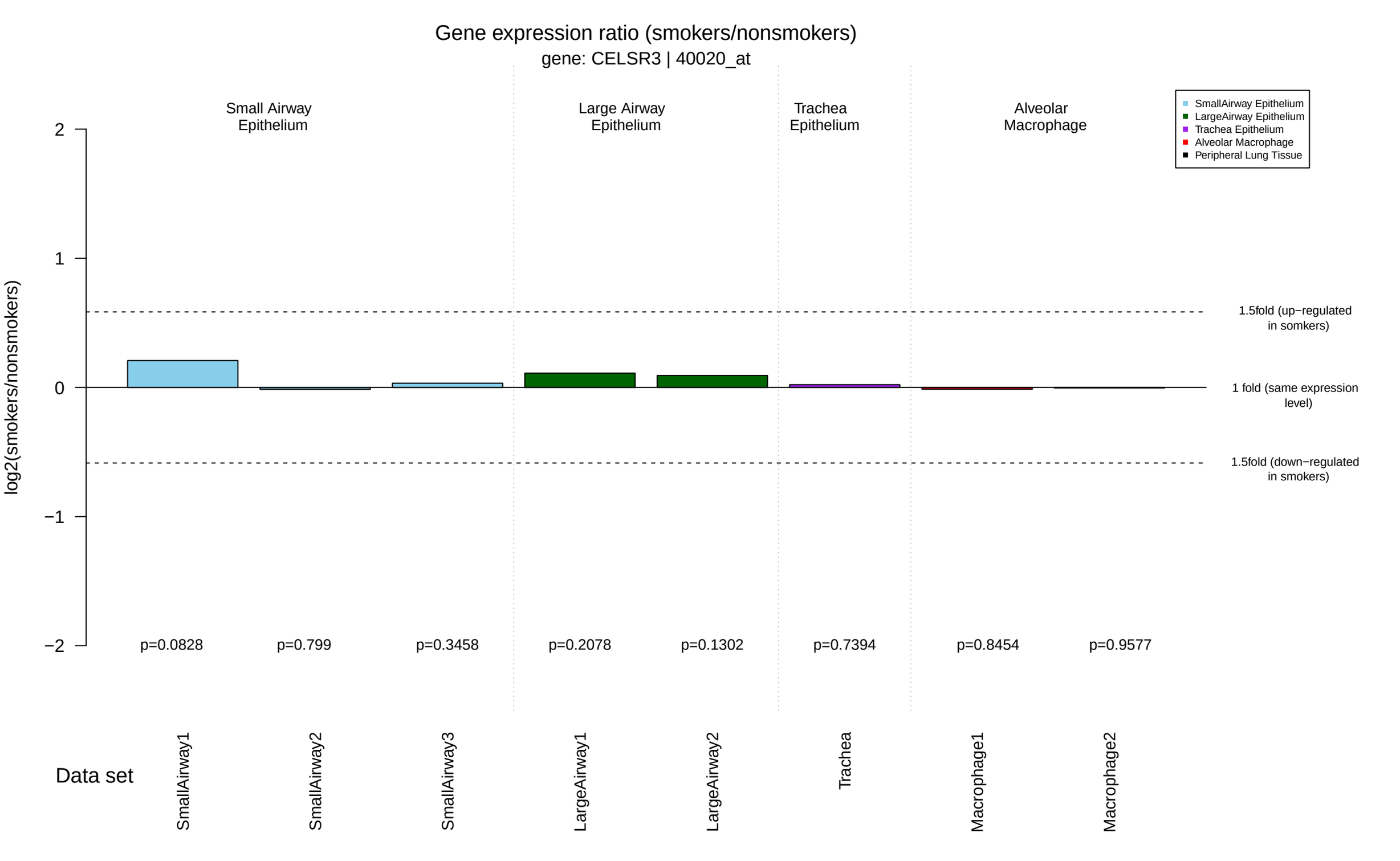
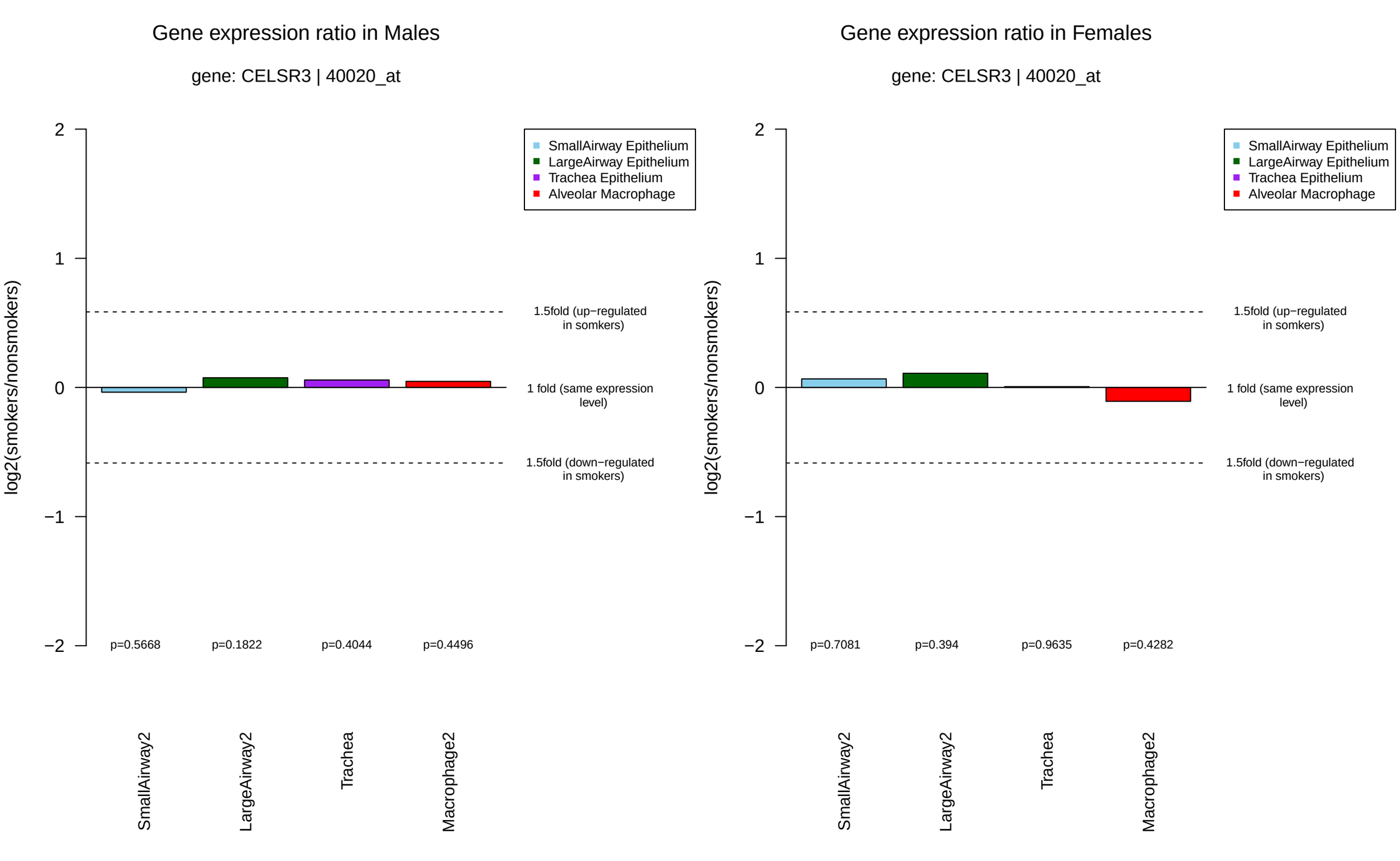
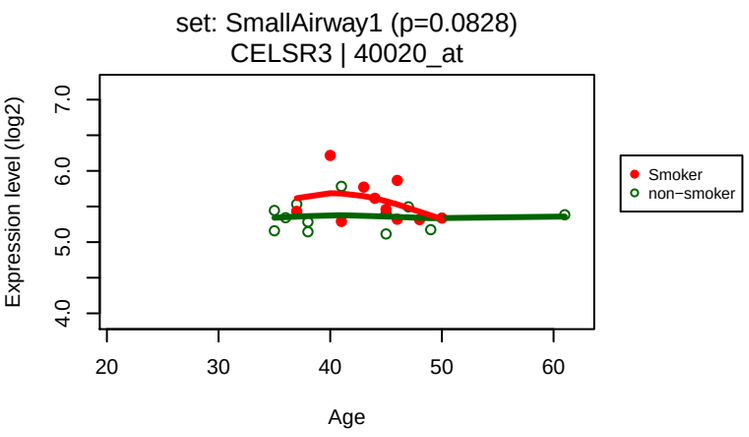
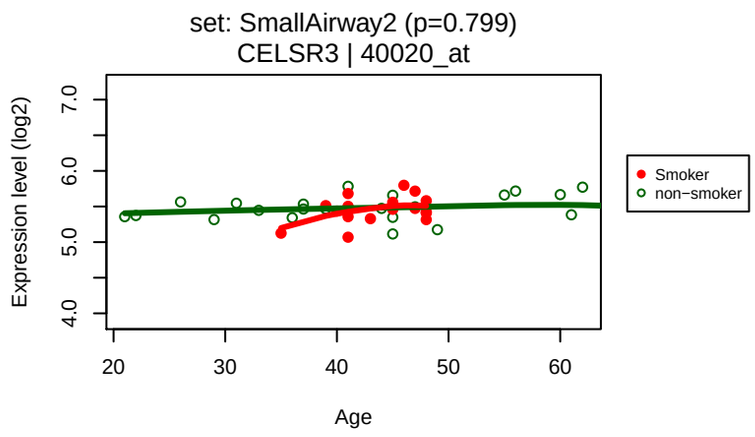
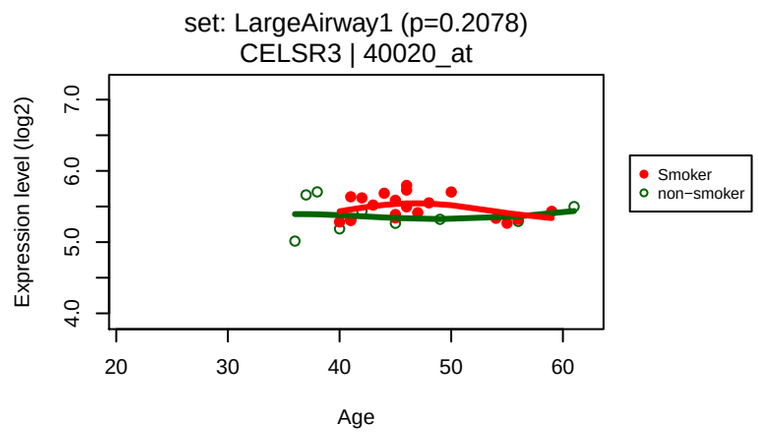

Gene expression of smokers and nonsmokers
gene: CELSR3 | 40020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0828
p=0.799
p=0.3458
p=0.2078
p=0.1302
p=0.7394
p=0.8454
p=0.9577
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

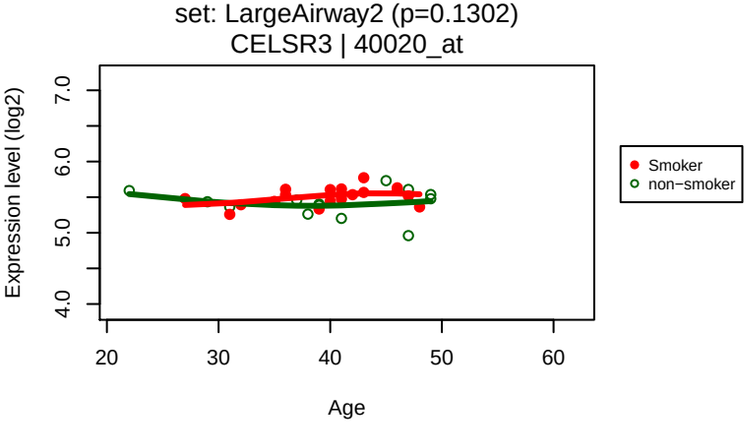

Gene expression ratio (smokers/nonsmokers)
gene: CELSR3 | 40020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0828
p=0.799
p=0.3458
p=0.2078
p=0.1302
p=0.7394
p=0.8454
p=0.9577
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

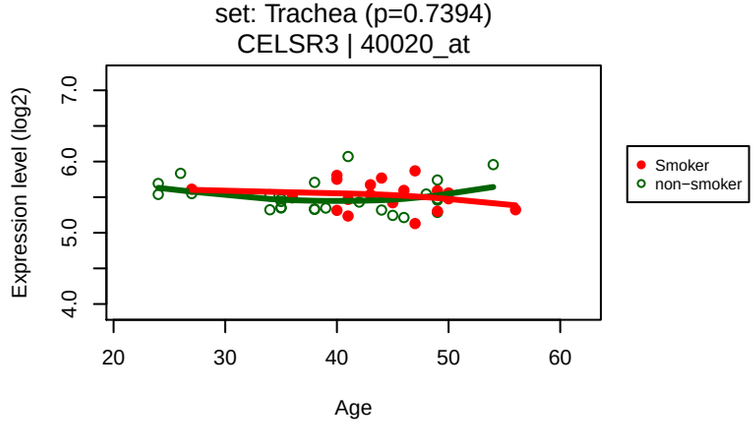

Gene expression ratio in Males
gene: CELSR3 | 40020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5668
p=0.1822
p=0.4044
p=0.4496
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CELSR3 | 40020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7081
p=0.394
p=0.9635
p=0.4282
SmallAirway2
LargeAirway2
Trachea
Macrophage2

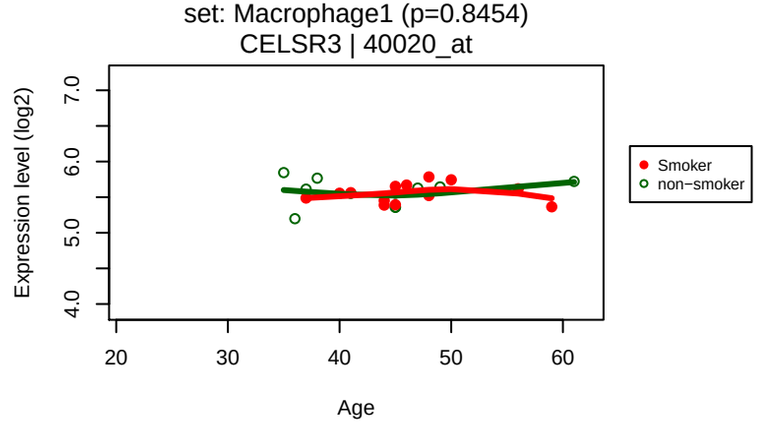

set: SmallAirway1 (p=0.0828)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

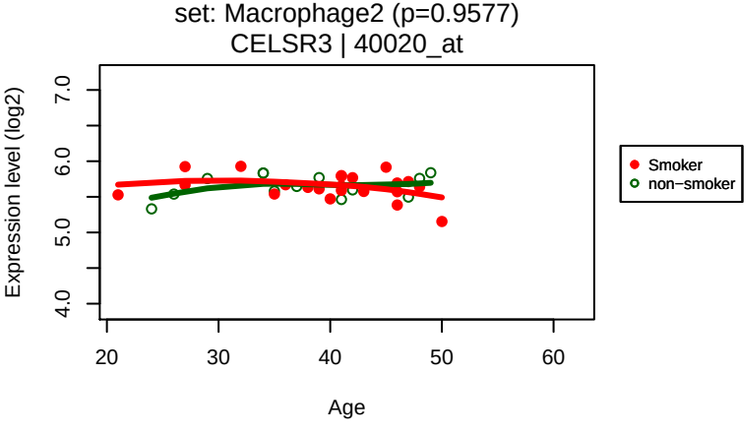

set: SmallAirway2 (p=0.799)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

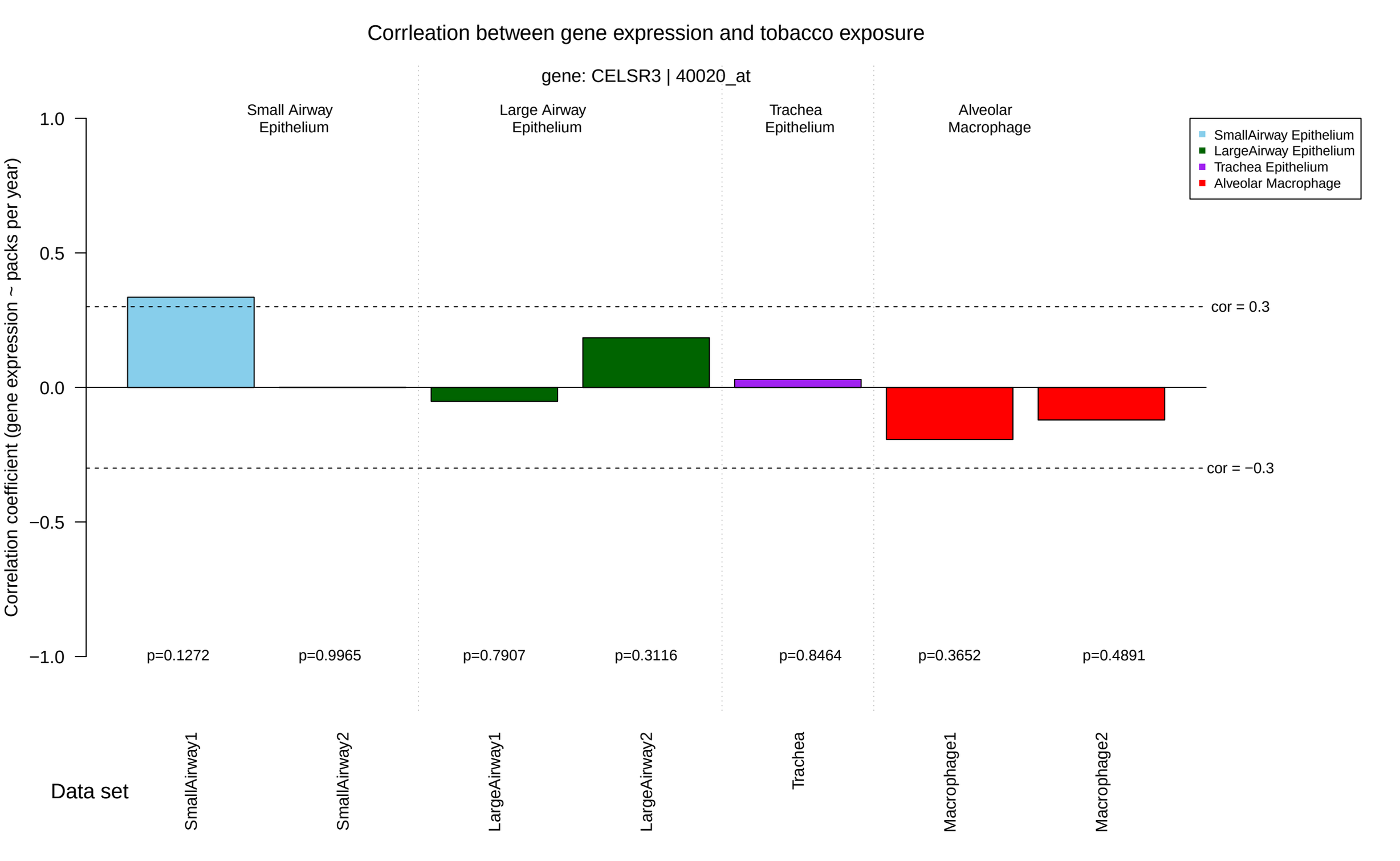

set: LargeAirway1 (p=0.2078)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1302)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7394)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8454)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9577)
CELSR3 | 40020_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CELSR3 | 40020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1272
p=0.9965
p=0.7907
p=0.3116
p=0.8464
p=0.3652
p=0.4891
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2